

The Missing Middle: A Systematic Review of Machine Learning for Municipal-Level Residential Energy Forecasting

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Abstract—Municipal-level energy forecasting is essential for local energy policy, yet it remains almost entirely absent from the machine learning literature. This paper presents a systematic review of 26 peer-reviewed studies published between 2020 and 2026, examining how machine learning methods perform across four spatial scales: building, neighborhood, municipality, and nation. Each empirical study is coded along four dimensions (spatial scale, temporal resolution, modelling approach, and best-performing model family) to enable a structured cross-scale comparison. The review finds that tree-based ensemble models, particularly Random Forest and XGBoost, perform best in 14 of the 22 empirical studies. Feature importance changes with temporal resolution: outdoor temperature and lagged consumption dominate at short intervals, while building characteristics and socioeconomic variables become the primary drivers at annual resolution. Only three studies operate at the municipal scale, and only one targets annual gas consumption; it excludes climate and building features and provides no interpretability analysis. We formally identify this as the ‘missing middle’: no existing study has applied a feature-complete model combining climate, building, and socioeconomic variables to annual municipal gas forecasting, and none has used SHAP analysis to explain which features drive consumption differences between municipalities. The Dutch setting, with CBS, BAG, and KNMI data available, is well-positioned to close this gap.

Index Terms—natural gas consumption, municipal-level forecasting, machine learning, systematic review, residential energy, feature importance, energy transition

I. INTRODUCTION

Accurate forecasts of natural gas consumption at the municipal level are essential for local energy policy. Local governments need these forecasts to plan insulation programs, heat pump installations, and district heating networks. In the Netherlands, the national goal to remove natural gas from residential heating by 2050 puts significant responsibility on municipalities [1]. Without precise data at this scale, it is difficult to prioritize interventions or to measure the success of local Dutch energy transition strategies.

The methodology for energy forecasting has evolved significantly in recent years. Traditional statistical methods (such as ARIMA and linear regression) are increasingly being replaced by machine learning techniques [2]–[4]. Tree-based

ensemble models (like Random Forest and XGBoost) have become common because they can handle the complex, non-linear relationships found in residential energy data [5], and tools such as SHAP (SHapley Additive exPlanations) values make their predictions interpretable for policymakers [6], [7]. Kernel methods, such as Support Vector Regression, have a competitive performance on smaller datasets [8], while deep learning strategies, such as Long Short-Term Memory (LSTM) networks, can deal with complex temporal variables but require substantially more data and computational resources [9].

At the building level, studies rely on detailed sensor data that is rarely available for many locations at once [9], [10]. At the national level, models use broad economic indicators that hide local differences [8], [11]. The municipal scale sits between these two extremes, but it is almost entirely absent from the literature. A review of the recent literature finds three studies at the municipal scale. But, only one examines annual natural gas consumption: Gawel and Paliński (2024) [12], who study Polish municipalities using 20 years of historical data. Their model includes lag features, population, and dwelling counts, but leaves out temperature, building characteristics, and income data. It also provides no feature importance analysis, so it cannot explain why municipalities differ in their consumption. The other two municipal studies [13], [14] use daily data, which makes them better suited for short-term operations than for the annual planning that energy transition policy requires. None of these studies match the Dutch context, where the goal is not just to forecast consumption accurately, but also to understand what drives differences between municipalities so that interventions can be targeted. There is currently no validated forecasting methodology for this specific scale and context.

This paper addresses that gap by systematically reviewing 26 recent studies (2020–2026). This analysis leads to two main contributions. First, we provide the first cross-scale synthesis of model performance for residential energy forecasting, establishing that tree-based ensembles are the dominant model family at building and national scales. Second, we formally identify and name a research gap at the municipal scale —

which we call the ‘missing middle’ — and show that the gap is twofold: no study has applied a feature-complete model combining climate, building, and socioeconomic variables to annual municipal gas forecasting, and no study has used interpretability methods to explain which of those features drive consumption differences between municipalities.

II. LITERATURE REVIEW

This review follows a systematic approach to identify and analyze current trends in residential energy forecasting using machine learning. The research process started with a structured search across four major academic databases: Scopus, Web of Science, IEEE Xplore, and Google Scholar, with the search last updated in early 2026. These databases were selected because they together provide broad coverage of engineering, computer science, and interdisciplinary research relevant to this topic. Only English-language publications were considered. The search string used for the title, abstract, and keyword search was:

("natural gas consumption" OR "residential energy consumption" OR "heating energy demand" OR "domestic energy consumption" OR "domestic energy prediction") AND ("forecasting" OR "prediction" OR "machine learning") AND ("municipal" OR "residential buildings" OR "household" OR "county" OR "neighbourhood" OR "postal code")

The string is split in three parts connected by AND operators: energy type, forecasting method, and spatial scope. Within each part, OR operators broaden the search to account for different terminology used across the literature.

The scope was limited to peer-reviewed publications from 2020 to early 2026 to ensure that the studies reflect the most recent developments in machine learning methods for energy forecasting. Studies were selected based on three primary inclusion criteria. First, the paper must employ a machine learning or statistical forecasting method. Second, the target variable must be residential energy consumption, specifically focusing on natural gas or electricity. Third, the study must report quantitative evaluation metrics, such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), or the Coefficient of Determination (R^2). Studies were excluded if they focused solely on commercial or industrial energy use, reported only qualitative findings without quantitative metrics, or if the full text was unavailable.

The selection process involved a multi-stage screening sequence, as illustrated in Figure 1. An initial search returned 4,277 records across all four databases. After removing duplicates, 2,418 titles and abstracts were screened for relevance, with 2,273 records excluded at this stage. The remaining 145 papers underwent a full-text assessment, resulting in a final set of 26 records comprising 22 empirical studies and 4 comprehensive literature reviews.

To enable a structured comparison, each empirical study was coded along four dimensions: spatial scale, temporal resolution, modelling approach, and best-performing model

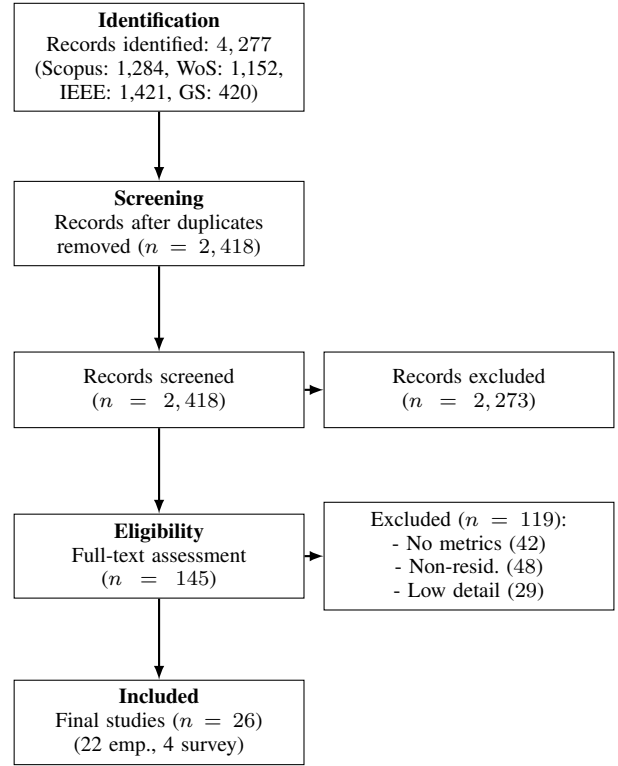


Fig. 1. PRISMA flow diagram for the selection of studies [15].

family. These dimensions were chosen because they capture the methodological choices most relevant to the municipal forecasting context. The spatial scale indicates the geographic level of aggregation, from individual buildings to national totals. The temporal resolution captures the frequency of the data (hourly, daily, monthly, or annual). The modelling approach looks between global models trained on pooled data from all locations and local models fitted to individual locations. The best-performing model family identifies which algorithm achieved the highest accuracy in the study. Where studies reported multiple metrics, R^2 and MAPE were prioritized for comparison.

III. SYNTHESIS

Table I provides a summary of the 22 empirical studies analyzed in this review. These studies are categorized using the four dimensions described in Section II. The research is split across four spatial scales: 10 studies focus on individual buildings, 3 study neighborhood clusters, 3 study municipalities, and 6 examine national data. As shown in Figure~3, the research is heavily concentrated at the building scale, with coverage thinning sharply in the middle before recovering at the national level. As shown in Figure~2, tree-based models are the most successful, as they are the best-performing family in 14 of the 22 studies. Deep learning and kernel methods each lead in 4 studies. In terms of modeling approach, 11 studies use a global model trained on pooled datasets, while 11 use local models for individual locations. Regarding temporal

TABLE I
SUMMARY OF REVIEWED STUDIES

Author	Year	Region	Scale	Energy type	Frequency	Approach	Best model	Best result	XAI
Rahmatov et al.	2024	Asia	Building	Natural gas	Monthly	Local	Deep Learning	RMSE = 19,651	—
Olu-Ajayi et al.	2022	Europe	Building	Electricity	Annual	Global	Deep Learning	$R^2 = 0.95$	—
Bassi et al.	2021	North America	Building	Electricity	Hourly	Local	Tree-based	$R^2 = 0.99$	—
Elhabyb et al.	2024	North America	Building	Electricity	Hourly	Local	Tree-based	$R^2 = 0.998$	—
Rambabu et al.	2022	Europe	Building	Electricity	Hourly	Local	Tree-based	$R^2 = 0.75$	—
Tasić et al.	2026	Europe	Building	Natural gas	Annual	Local	Tree-based	$R^2 = 0.99999$	—
Vonitsanos et al.	2024	North America	Building	Electricity	Hourly	Local	Kernel Methods	$R^2 = -0.05$	—
Vaisi et al.	2025	Asia	Building	Natural gas	Annual	Global	Tree-based	$R^2 = 0.90$	—
Kemper et al.	2025	Europe	Building	Natural gas	Hourly	Local	Kernel Methods	MASE = 0.26	—
Navale et al.	2025	Global	Building	Electricity	Hourly	Global	Deep Learning	$R^2 = 0.97$	—
Colverd et al.	2025	Europe	Neighborhood	Natural gas	Annual	Global	Tree-based	$R^2 = 0.93$	—
Jafarigorzin et al.	2025	Europe	Neighborhood	Electricity	Annual	Global	Tree-based	MAE = 2.1%	—
Fan et al.	2025	Asia	Neighborhood	Electricity	Annual	Global	Tree-based	$R^2 = 0.667$	SHAP
Panek & Wlodek	2022	Europe	Municipal	Natural gas	Daily	Local	Tree-based	MAPE = 1.61%	—
Mittakola et al.	2024	North America	Municipal	Natural gas	Daily	Global	Tree-based	—	SHAP
Gawel & Paliński	2024	Europe	Municipal	Natural gas	Annual	Global	Deep Learning	MAPE = 7.3%	—
Cui et al.	2024	North America	National	Electricity	Annual	Global	Tree-based	$R^2 = 0.78$	SHAP
Mohammadizazi & Bilec	2020	North America	National	Electricity	Annual	Global	Tree-based	$R^2 = 0.72$	—
Eshita et al.	2025	North America	National	Natural gas	Monthly	Global	Tree-based	$R^2 = 0.998$	SHAP
Singh et al.	2023	North America	National	Natural gas	Annual	Local	Kernel Methods	MAPE = 2.7%	—
Fariás et al.	2025	South America	National	Electricity	Monthly	Global	Tree-based	$R^2 = 0.96$	—
Gao & Shao	2021	Asia	National	Natural gas	Annual	Local	Kernel Methods	MAPE = 0.51%	—

resolution, 11 studies use annual data, 6 use hourly, 3 use monthly, and 2 use daily; notably, only one of the three municipal studies uses annual data, which is the resolution most relevant for long-term energy planning. The following subsections present findings by spatial scale, covering model performance and feature importance at each level.

A. Individual Building Studies

Research at the individual building level is well-developed, with ten of the 22 reviewed studies focusing on this scale. The studies predict gas consumption for single homes or specific building complexes. They show a difference between different model families. Deep learning models, such as Deep Neural Networks (DNN) and Long Short-Term Memory (LSTM) networks, often perform best when large datasets are available. For example, Olu-Ajayi et al. [16] achieved an R^2 of 0.95 using a DNN on 5,000 UK buildings. Navale et al. [9] reached an R^2 of 0.97 using autoencoders on the large-scale ASHRAE dataset. These results suggest that deep learning is good at finding complex patterns when there is plenty of historical data.

However, deep learning does not always perform best. On smaller or more specific datasets, tree-based and kernel methods are often just as accurate [17]. Vaisi et al. [18] reported that XGBoost and Random Forest both reached an R^2 of 0.90 for residential buildings in Iran, and Bassi et al. [10] confirmed that gradient boosting approaches like XGBoost are strong performers at the building scale. Kernel methods can also outperform deep learning when data is limited: Kemper et al. [19] found that Kernel Ridge Regression was more accurate than both deep learning and time-series models for a 66-unit residential complex in Germany. These results suggest that

model complexity does not guarantee better performance. But, data quality matters just as much as model choice. Vonitsanos et al. [20] showed that SVM and Random Forest both failed to beat a simple baseline, showing that that even well-established models break down on noisy or inconsistent data.

Beyond model choice, the literature consistently identifies physical characteristics as the strongest predictors for the building level [21]. The total floor area is the most important variable across multiple studies, holding across different climates and building types [16], [18]. Building age and the number of rooms also matter, though they rank below floor area. These findings suggest that the size and physical structure of a building is responsible for its baseline energy demand.

A major limitation of building-level research is that results rarely generalize beyond their original context. Most studies focus on a single building or a small group [22], and models trained on specific locations are not validated for different building types or climates. Elhabyb et al. [23], for instance, achieved an R^2 of 0.998 on educational buildings, a result that cannot transfer directly to residential settings. This is further visible in extreme cases: Tasić et al. [24] reported an R^2 of 0.99999, a score that suggests the model memorized its training data rather than learning transferable patterns. A second limitation is temporal resolution: most building-level studies use hourly or monthly data, but annual resolution is what matters for long-term energy planning. Of the ten building studies reviewed, only three use annual data [16], [18], [24]. This means that the majority of building-level findings are better suited to short-term operations than to the kind of strategic planning that energy transition policy requires. Together, these limitations mean that building-level models, however accurate locally, are too narrow to support

regional energy policy planning.

B. Neighborhood-level Studies

At the neighborhood level, the focus shifts from individual buildings to clusters. These are usually defined by postal codes or small administrative zones. This scale is important for local planning because it provides detail for specific areas while keeping the data manageable. Recent studies at this level show that tree-based models are very accurate at capturing the energy behavior of these groups.

One of the most relevant studies at this scale was by Jafarigorzin et al. [25], who studied the Netherlands using four-digit and six-digit postal codes. Their work confirmed that buildings and demographic features matter much more than climate at this level. By using XGBoost, they found that the previous year's consumption was the most powerful predictor. It explained over 97% of the results. When they removed this historical data, the number of households and the average floor area became the main drivers. Their research also showed an interesting point: even though individual homes are strongly affected by temperature, this relationship often disappears when looking at a whole neighborhood. This indicates that weather causes daily changes, but the buildings themselves determine the basic annual energy use.

Two studies have useful methodological notes for neighborhood-level forecasting. Colverd et al. [26] demonstrated that automated machine learning tools can reach an R^2 of 0.93 for gas consumption at the postcode level in England and Wales, suggesting that strong results are achievable without manual model tuning. Fan et al. [27] looked at a different practical constraint: in regions where billing data is unavailable, GIS-derived building characteristics can serve as a substitute. By identifying representative building prototypes from urban shape data, their XGBoost model could map energy use by Chinese neighborhoods without any direct consumption records. These studies show that neighborhood-level forecasting is methodologically flexible, adapting to different data availability conditions. And while consistently relying on physical building features as the core input.

For policymakers, the key takeaway is that annual energy consumption at the neighborhood level is highly stable from year to year, reflecting the slow pace of change in the built environment rather than fixed behavioral patterns. If historical data is available, the lagged features alone explain most of the variance. When it is not, physical property indicators such as floor area and household count serve as reliable substitutes. Planners can use property registries to identify high-consumption neighborhoods and prioritize them for retrofit programs.

C. Municipal-level Studies

The municipal scale is a critical middle ground, but it is less explored than individual buildings or the national level. Building-level models are precise, and national models show broad trends. However, the municipal level is where local energy policy and investment decisions are made. Research

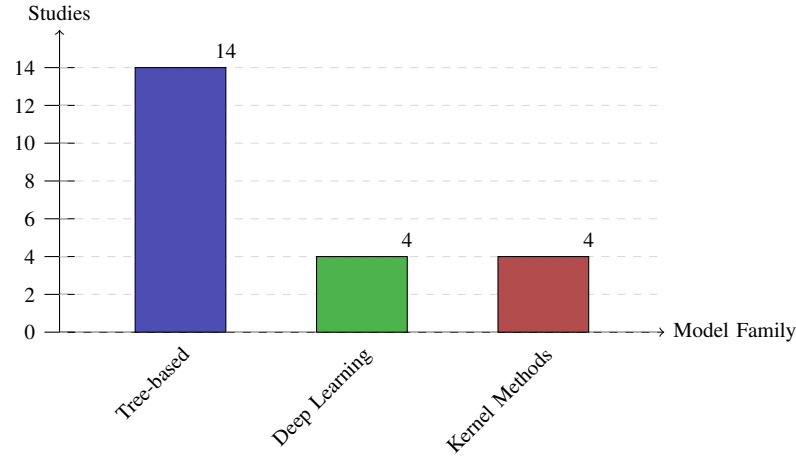


Fig. 2. Distribution of best-performing model families across the 22 reviewed empirical studies.

at this scale is essential to understand how local geography, demographics, and urban planning affect gas use. However, the small number of studies shows a gap in the literature, especially for long-term annual planning. None of the three studies reviewed here address the Dutch context, where the goal is not only accurate forecasting but also understanding what drives consumption differences between municipalities so that energy transition interventions can be prioritized.

One of the few detailed studies at this scale was by Panek and Wlodek [13]. They looked at natural gas demand for a city in Poland. Their work focused on daily patterns and compared several models. They found that Random Forest was the most accurate, with a Mean Absolute Percentage Error (MAPE) of just 1.61% for same-day forecasts. The study demonstrated that at the city level, two main factors are responsible for short-term consumption: outdoor temperature and the previous day's gas use. Temperature had a very strong negative correlation ($r = -0.91$) with gas demand. Other weather variables helped slightly, but factors like cloud cover or the specific day of the week had almost no effect. This research shows that short-term municipal demand is easy to predict using temperature and historical data. But these models often miss the structural reasons why energy is used.

To better understand these structural reasons, Mittakola et al. [14] analyzed 1,000 U.S. counties. They developed a model to define the relationship between temperature and gas use for each county. They found two specific indicators: the temperature where heating begins and how fast consumption increases as it gets colder. Using models like Random Forest and CatBoost, they found that building age and socioeconomic factors are the main drivers. For example, counties with many homes built between 1950 and 1999 were much more sensitive to temperature changes. The study also found a problem with energy poverty. Wealthy counties used 57% less gas per household than poorer counties, even with the same climate. This suggests that at the municipal scale, income and

employment are just as important as insulation.

Gawel and Paliński [12] addressed the challenge of annual municipal forecasting. They studied 1,851 municipalities in Poland using 20 years of data. They compared “local” models for individual towns with “global” models trained on all data at once. Their results showed that global machine learning models performed much better than traditional local methods. The local models struggled with short and noisy data. The global model achieved a MAPE of 7.3% for municipal predictions. A key finding was that linear regression failed at this scale, with errors over 30%. This confirms that the relationship between population, building counts, and gas use is complex and non-linear. However, like other studies at this scale, their model relied mostly on demographic and historical data. It left out building characteristics and climate indicators that are important at smaller scales. Furthermore, their study includes no feature importance analysis, which means it cannot explain which factors drive the differences in consumption between municipalities.

Together, these three studies show that machine learning works at the municipal scale, but each leaves a critical part of the problem unanswered. Short-term models achieve low error rates, but only because they rely on temperature and lagged consumption — they cannot explain why municipalities differ structurally. Mittakola et al. identify building age and income as drivers of county-level differences, but they model temperature sensitivity parameters rather than annual consumption levels, and their U.S. context involves different data structures than European municipal registries. Gawel and Paliński demonstrate that global ML models outperform local ones for annual municipal forecasting, but their model uses only demographic and historical data. Because it excludes climate variables and building characteristics, and includes no feature importance analysis, it cannot explain which factors drive consumption differences between municipalities, which is precisely what energy transition planning requires. This reveals a twofold gap. First, no study has applied a feature-complete model combining climate, building, and socioeconomic variables to annual municipal gas forecasting. Second, no study has used interpretability methods such as SHAP analysis to explain which features drive consumption differences at this scale. Both gaps are especially relevant for the Netherlands, where CBS and BAG registers make such a study feasible, but no validated methodology yet exists.

D. National-level Studies

National-level studies focus on predicting energy use for entire countries or large regions. These studies usually look at long-term trends over several decades. Because they cover such a large scale, they often use macro indicators like population and economic growth. The research shows that machine learning is very effective for finding broad patterns, with tree-based models being the most common choice.

Tree-based models like Random Forest, XGBoost, and CatBoost consistently deliver high accuracy. For example, research in Brazil showed that these models could reach an R^2

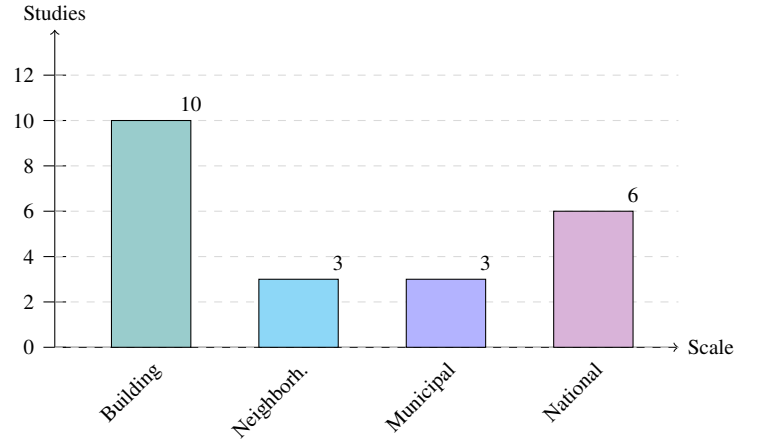


Fig. 3. Distribution of reviewed empirical studies by spatial scale, illustrating the research gap at the municipal level.

above 0.95 for residential electricity [5]. In the United States, several studies confirmed that ensemble methods generally perform better than traditional linear regression [7], [28]. While deep learning is often used for buildings, it is less common for national annual predictions. This is because the datasets are often too small for complex neural networks. Instead, Support Vector Machines (SVM) are a strong choice for small annual datasets. Studies in China and the U.S. showed that SVM can achieve very low error rates when predicting gas use with just a few decades of data [8], [11].

The importance of different features changes at the national scale. For long-term forecasts, socioeconomic variables like GDP per capita, total population, and urbanization rates are the strongest predictors [8], [11]. These drivers explain most of the growth in energy use over many years. However, when research uses more detailed data—like monthly consumption—other features matter more. Climate variables, like heating degree days and outdoor temperature, are critical for capturing seasonal changes in energy use. Building characteristics also play a major role. Studies in the U.S. found that total floor area and the type of heating fuel were the most significant drivers for both apartments and single-family houses [7]. Across these studies, the most important features shift with temporal resolution: climate variables explain shorter-term variation, while building characteristics and socioeconomic factors drive long-term annual trends.

IV. DISCUSSION AND LIMITATIONS

Across all four spatial scales, tree-based ensemble models emerge as the dominant model family, performing best in 14 of the 22 empirical studies reviewed. Residential energy data is tabular, non-linear, and rarely large enough to train deep neural networks reliably. These are all conditions that favor tree-based methods. Deep learning remains competitive at the building scale when hourly sensor data is available, and kernel methods hold up for small annual datasets at the national scale, but

for annual municipal forecasting, tree-based ensembles are the clear starting point. The synthesis also reveals a consistent shift in feature importance by temporal resolution: outdoor temperature and lagged consumption dominate at short intervals, while building characteristics and socioeconomic variables become the primary drivers at annual resolution.

The absence of municipal-scale research reflects a structural data problem. Building-level studies rely on sensor or metering data; national studies use macro statistics published by statistical agencies. The municipal scale requires combining consumption records, building databases, demographic data, and climate inputs from different institutions, and this integration step is demanding enough to discourage research. This data challenge has real policy costs. In the Netherlands, municipalities are legally responsible for local energy transition planning under the Climate Agreement [1]. The absence of a validated forecasting methodology at this scale is not just a research gap but a practical obstacle for policymakers.

The Dutch context is nonetheless well-suited to close this gap. CBS publishes annual consumption data per municipality, the BAG register provides building characteristics for every address, and KNMI supplies climate records including heating degree days. Together, these sources make it possible to build the feature-complete dataset that no existing municipal study has assembled. Based on the findings of this review, a global model trained on all municipalities simultaneously is preferable to local models, XGBoost or Random Forest are the appropriate candidates, and SHAP analysis should be included from the outset to explain which features drive consumption differences between municipalities.

This review has several limitations. The search covered only English-language publications from a limited set of databases, so some relevant studies may have been missed. With only 26 papers in the final set, the observed patterns (like the dominance of tree-based models) reflect trends rather than firm conclusions. Coding each study along four dimensions also involved judgment calls by the author. And comparing results across studies is difficult because no two studies use the same dataset or evaluation setup.

V. CONCLUSION

This review formally identifies a twofold research gap at the municipal scale — the ‘missing middle’. No existing study has applied a feature-complete model combining climate, building, and socioeconomic variables to annual municipal gas forecasting, and none has used SHAP analysis to explain which features drive consumption differences between municipalities.

The evidence across 22 empirical studies is consistent. Tree-based ensembles outperform other model families in 14 of them. Feature importance shifts by temporal resolution: temperature and lagged consumption dominate at short intervals, while building characteristics and socioeconomic variables drive annual variation. The three municipal studies each address part of the problem, but none is feature-complete or interpretability-focused.

The Dutch context is well-positioned to close this gap. CBS, BAG, and KNMI together provide the data needed for a feature-complete annual municipal dataset. Based on this review, a global XGBoost or Random Forest model with SHAP analysis is the right starting point.

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